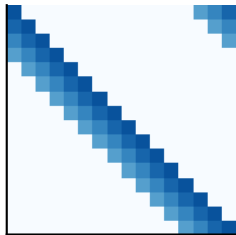
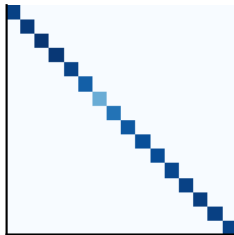


(a) channel with CP: circulant



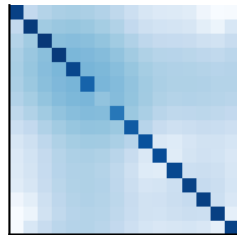
each row = previous, shifted

(b) $\mathbf{F}\mathbf{H}_c\mathbf{F}^H$: diagonal



off-diagonal energy: zero

(c) no CP: leakage (ICI)



off-diagonal energy -15 dB

dB

0
-10
-20
-30
-40